Polish-Japanese Academy of Information Technology

Bachelor's engineering thesis

Short-Term Forecasting of Martian Surface Weather Using Machine Learning

PRO3, Task 07: Chapter 7, Results and discussion

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7 Results and discussion

This chapter reports the outcomes of the three experiments defined in Chapter 6 and interprets them. It is organized by research question: Section 7.1 covers the accuracy of the four model classes at Jezero Crater (RQ1), Section 7.2 the value of the secondary signals (RQ2) and Section 7.3 the transfer of models between the three sites (RQ3). Each section states briefly what was run, shows the results and discusses what they mean. Section 7.4 then relates the results to the aims of the thesis and to the literature and states the limitations, strengths and weaknesses of the work, and Section 7.5 summarizes the chapter.

The numbers come from the runs of 27 October 2026 on the corrected data (Chapter 4) and are computed from the forecasts stored with each run (Chapter 5). For RQ1 and RQ2, 132 models were fitted on the MEDA 5-minute table: twelve temporal convolutional networks (TCN) and, for each of the two feature sets and five seeds, twelve gradient-boosted decision tree ensembles (GBDT); the two reference forecasts need no fitting. All of them were scored on the same 236,127 test forecasts. A further 60 GBDTs were fitted on 12 November 2026 for one additional check that the results of RQ1 called for (Section 7.1.3). For RQ3, 48 GBDTs were fitted on the hourly tables of the three stations. Unless stated otherwise, an error is the root mean square error (RMSE) on the test period, a GBDT number is the mean over seeds 1 to 5, and an interval is a 95 % interval of the sol-block bootstrap (Chapter 6), which is applied here to the skill scores of the single-site GBDT as well.

7.1 Single-site forecasting accuracy

7.1.1 Accuracy and skill of the four model classes

Table 7.1 and Figure 7.1 give the test RMSE of persistence, the typical day, the GBDT and the TCN for the three targets and four horizons, and Table 7.2 gives the mean absolute error (MAE) and the skill scores. The three targets behave differently. For air temperature, both learned models are far more accurate than both references at every horizon. For pressure, persistence is the most accurate forecast at every horizon and neither learned model, as designed in Chapter 6, matches it. For wind speed, the GBDT and the typical day are equally accurate, and the TCN is worse than both.

The MAE leads to the same conclusions as the RMSE. The order of two models changes only where the RMSE already places them within a few percent of each other: for air temperature at one hour, where the GBDT has the lower MAE and the TCN the lower RMSE, and for wind at 1 and 24 hours, where the typical day has a slightly lower MAE than the GBDT. The conclusions therefore do not rest on a few large errors. The choice of the reference, in contrast, changes the size of the skill considerably. Against persistence the learned models reach a skill of +0.87 for air temperature at six hours, but persistence is a weak opponent there, with an error of 23.4 K,

Table 7.1: Test RMSE of the four model classes at Jezero Crater

Target	Horizon (h)	Forecasts	Sols	Persistence	Typical day	GBDT	TCN
Air temperature (K)	1	35,496	222	5.35	4.51	2.10 ± 0.01	**2.07**
	6	27,123	222	23.43	4.47	**2.95** ± 0.04	2.96
	24	28,882	217	3.93	4.57	**2.02** ± 0.02	2.24
	48	22,411	213	6.86	4.50	**2.27** ± 0.01	2.87
Pressure (Pa)	1	35,410	222	**2.72**	36.17	5.75 ± 0.11	3.28
	6	27,004	222	**8.70**	36.38	13.78 ± 0.84	10.17
	24	28,794	217	**2.14**	36.04	6.03 ± 0.32	8.08
	48	22,333	213	**3.85**	36.12	7.99 ± 0.79	10.36
Wind speed ($m\,s^{-1}$)	1	2,027	37	1.65	1.33	**1.29** ± 0.01	1.56
	6	2,796	36	2.94	**1.33**	1.37 ± 0.01	1.61
	24	2,232	34	1.61	1.26	**1.25** ± 0.01	1.90
	48	1,619	29	1.71	**1.24**	**1.24** ± 0.02	1.38

Forecasts: scored issue times of the test period (sols 1389–1619; wind 267–312); sols: test sols that contain them. GBDT: mean and standard deviation over seeds 1–5; TCN: one run. Bold: lowest error of the row; differences below the run-to-run variation discussed in the text are ties.

Source: own elaboration based on the stored forecasts of the final runs (Chapter 5).

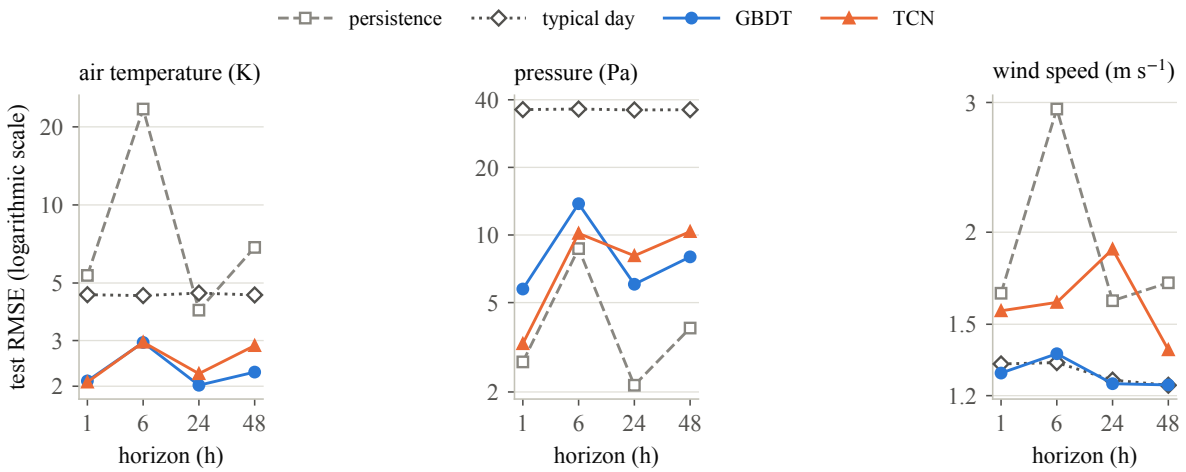
because six hours lead to the opposite side of the diurnal ramp. Against the typical day, which is the better reference in that row, the skill is +0.34. The typical day beats persistence for air temperature at 1, 6 and 48 hours and for wind at every horizon, and persistence beats the typical day only for air temperature at 24 hours and for pressure: its own skill against persistence is +0.16, +0.81, −0.16 and +0.34 for air temperature, +0.20, +0.55, +0.22 and +0.27 for wind and below −3 for pressure. Reporting the skill against the better of the two references, as Murphy recommended[1], is therefore necessary to avoid flattering the models.

7.1.2 Air temperature

Air temperature is the target on which machine learning pays off. The GBDT forecasts it with an error of 2.0–3.0 K at all four horizons, where the diurnal range is 52–61 K (Chapter 2), and its skill against the better reference is +0.34 to +0.53. The spread over the five seeds is at most 0.04 K, far below any difference between model classes. The error does not grow steadily with the horizon: it is lowest at 1 and 24 hours, when the current value or the value one sol earlier is informative, and highest at 6 hours. The feature importances of the GBDT show how the model adapts; Table 7.3 lists the three largest shares of selected models, and the sums over groups of features quoted in the text come from the full list stored with the figures of this chapter. At one and six hours, 84 and 93 % of the gain comes from the solar angles and the time of sol, that is, from the position of the forecast on the diurnal cycle; at 24 and 48 hours, 67–88 % comes from the recent history of the air temperature itself. The model thus places the forecast on the cycle first and corrects it with the current state, a combination that neither reference can make.

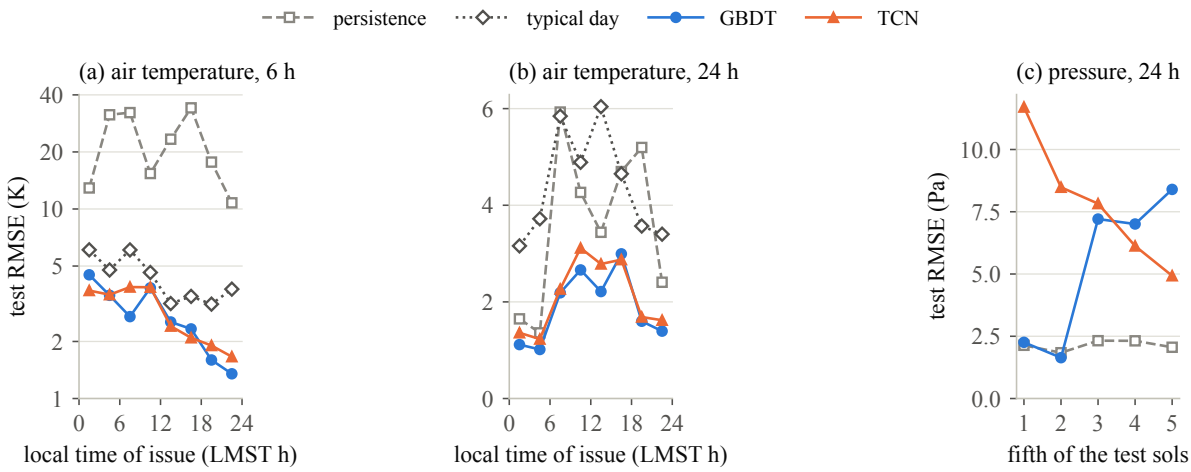
[1]Cf. A. H. Murphy. “Climatology, Persistence, and Their Linear Combination as Standards of Reference in Skill Scores”. In: *Weather and Forecasting* 7.4 (1992), pp. 692–698.

Figure 7.1: Test RMSE of the four model classes by forecast horizon



Source: own elaboration based on the stored forecasts of the final runs (Chapter 5).

Figure 7.2: Test RMSE by local time of the issue time and by part of the test period



Source: own elaboration based on the stored forecasts of the final runs (Chapter 5).

Figure 7.2 locates the errors within the sol. At six hours (panel a), persistence errs by 31–34 K when the forecast is issued between 3 and 9 or between 15 and 18 LMST, across sunrise or sunset, whereas the learned models stay below 4.6 K in every part of the sol. At 24 hours (panel b), where persistence is strongest, the GBDT still reduces its error by 36–69 % for forecasts issued between 6 and 24 LMST. Their own errors follow the boundary layer described in Chapter 2: they are lowest in the stable night, about 1.0–1.4 K for forecasts issued between 0 and 6 LMST, and highest in the convective daytime, 2.2–3.1 K between 9 and 18 LMST. At one hour the same contrast is stronger for the GBDT, 0.8 K for forecasts issued between 0 and 3 LMST against 2.2–2.9 K between 6 and 18 LMST. A large part of the remaining daytime error is thus turbulent variability within the hour, which no model fed with 5-minute means can be expected to forecast. Over the test period the errors are stable: the 24-hour error of the GBDT lies between 1.92 and 2.11 K in each fifth of the test sols.

Table 7.2: Test MAE and skill scores of the learned models

		MAE				Skill against persistence		Skill against the better reference	
Target	H (h)	Pers.	Typ. day	GBDT	TCN	GBDT	TCN	GBDT	TCN
Air temperature (K)	1	4.18	3.71	1.53	1.60	+0.61	+0.61	+0.53	+0.54
	6	20.62	3.66	2.25	2.31	+0.87	+0.87	+0.34	+0.34
	24	3.01	3.75	1.44	1.68	+0.49	+0.43	+0.49	+0.43
	48	5.41	3.65	1.68	2.25	+0.67	+0.58	+0.50	+0.36
Pressure (Pa)	1	2.25	35.10	4.32	2.53	−1.11	−0.21	−1.11	−0.21
	6	7.26	35.32	12.27	9.37	−0.58	−0.17	−0.58	−0.17
	24	1.68	34.99	4.05	7.39	−1.82	−2.77	−1.82	−2.77
	48	3.04	35.09	5.69	9.65	−1.07	−1.69	−1.07	−1.69
Wind speed ($\mathrm{m\,s^{-1}}$)	1	1.25	1.00	1.00	1.27	+0.22	+0.05	+0.03	−0.18
	6	2.43	1.01	1.06	1.31	+0.53	+0.45	−0.03	−0.21
	24	1.19	0.93	0.96	1.54	+0.23	−0.17	+0.01	−0.51
	48	1.28	0.94	0.96	1.10	+0.27	+0.19	−0.00	−0.12

Skill: $1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{ref}}$ (Chapter 6). Better reference: the more accurate of persistence and the typical day in that row (Table 7.1).

Source: own elaboration based on the stored forecasts of the final runs (Chapter 5).

Figure 7.3 shows three test sols as an example. The six-hour forecasts of both learned models follow the measured air temperature through the whole cycle, including the maxima; the typical day reproduces the shape but lies a few kelvin too low at night, and persistence repeats the cycle six hours late.

The two learned models cannot be separated at the short horizons. The TCN is 1.0 % more accurate than the GBDT at one hour and 0.2 % less accurate at six hours, differences far below the run-to-run variation of the TCN of about 6 % (Chapter 6); these two cells are ties. At 24 and 48 hours the GBDT is better by 10.8 and 26.5 %, more than that variation. Table 7.4 adds a qualification. The errors of the GBDT are almost unbiased, within 0.25 K, whereas the TCN forecasts too warm at every horizon, by 0.5–1.4 K. The scatter of the TCN errors at six hours, 2.59 K, is nevertheless smaller than that of the GBDT, 2.94 K, so the network captures the variation at least as well and loses its advantage through an offset. The learning curves of Chapter 6 suggest why the longer horizons are harder for the network: its validation error turned upward after one to six epochs in eleven of twelve models, and for air temperature at one hour the first epoch was kept. With 84,000 to 122,000 training windows from one site, a network of 222,529 parameters fits the training period within a few passes, while the GBDT, which receives the history already condensed into lags and window statistics, needs less data to use it.

7.1.3 Pressure

For pressure the result is negative: persistence is the most accurate forecast at every horizon, with errors of 2.1–3.9 Pa at 1, 24 and 48 hours. The GBDT errs by 5.7–13.8 Pa and the TCN by

Table 7.3: Three most important features of selected GBDT models

Model	Features and their shares of the total gain
Primary inputs	
Air temperature, 1 h	solar zenith angle 71 %; mean air temperature of the last 6 h 10 %; cosine of the LMST hour 7 %
Air temperature, 6 h	solar azimuth angle 48 %; solar zenith angle 26 %; cosine of the LMST hour 10 %
Air temperature, 24 h	air temperature 5 min before 61 %; mean air temperature of the last hour 19 %; LMST hour 10 %
Air temperature, 48 h	mean air temperature of the last 6 h 52 %; solar zenith angle 20 %; air temperature 5 min before 10 %
Pressure, 1 h	seasonal baseline 69 %; mean pressure of the last 6 h 15 %; pressure 5 min before 5 %
Pressure, 24 h	seasonal baseline 69 %; mean pressure of the last 6 h 22 %; pressure 5 min before 4 %
Wind speed, 1 h	cosine of the LMST hour 32 %; solar zenith angle 14 %; solar azimuth angle 7 %
Wind speed, 24 h	air temperature 5 min before 30 %; mean air temperature of the last 6 h 14 %; solar zenith angle 9 %
With secondary signals	
Air temperature, 1 h	upward longwave flux 73 %; cosine of the LMST hour 8 %; solar zenith angle 6 %
Air temperature, 48 h	mean solar radiation of the last 6 h 59 %; mean upward longwave flux of the last 6 h 17 %; mean air temperature of the last 6 h 5 %

Gain: reduction of the training loss by the splits on a feature (Chapter 6). Boosters of seed 1, refitted for this table; their test forecasts equal those of the stored runs.

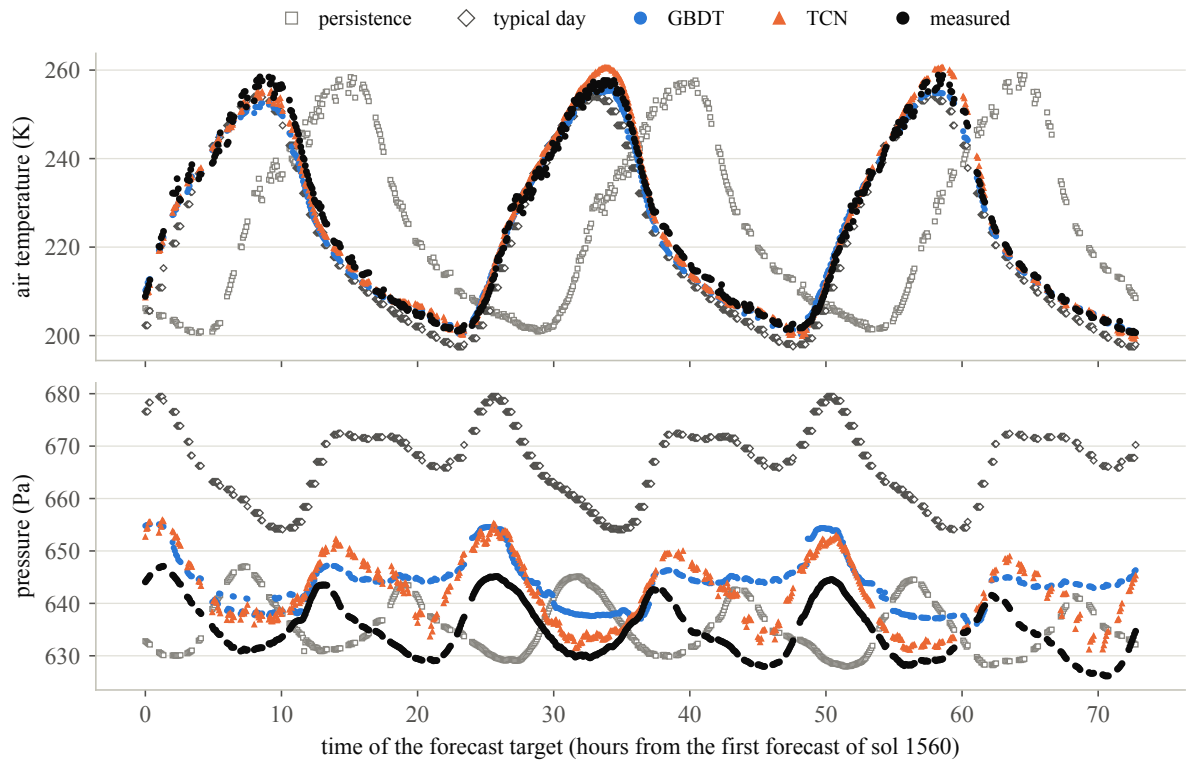
Source: own elaboration based on the feature tables (Chapter 4) and the experiment configuration.

3.3–10.4 Pa. The TCN is clearly better than the GBDT at 1 and 6 hours and worse at 24 and 48 hours, but it loses to persistence in every cell. The seed spread of the GBDT, 2–10 % of its error, is much larger than for the other targets, yet too small to change this ranking.

The error analysis shows that this failure has a specific cause and is not a limit of predictability. Three observations support this reading. First, most of the error of the learned models is an offset (Table 7.4). Both forecast too high at every horizon, by 3.7–12.2 Pa (GBDT) and 1.8–9.6 Pa (TCN), and the typical day, which adds a climatology to the seasonal baseline, is too high by 35 Pa. The direction and the size of the offset are those expected from the rover's climb to the crater rim: in the test period the measured pressure lies about 36 Pa below the baseline fitted on the training period, as the hydrostatic relation predicts for the gain in elevation (Chapter 4). The daily mean deviation from the baseline was between −27 and +23 Pa in the training period and between −48 and −21 Pa in the test period, so the models met inputs largely outside the range on which they were trained. Persistence carries no learned level and is unaffected. Without its offset, the scatter of the TCN errors (2.8, 4.0, 3.3 and 3.8 Pa) equals that of persistence at 1 and 48 hours, is less than half of it at 6 hours (8.7 Pa) and exceeds it only at 24 hours (2.1 Pa). The pressure panel of Figure 7.3 shows the same thing: both learned models reproduce the shape of the diurnal pressure cycle, which persistence shifts by six hours, but they draw it too high.

Second, the error is not uniform over the test period (Figure 7.2c). In the first two fifths of the test sols the GBDT forecasts pressure 24 hours ahead with errors of 2.3 and 1.6 Pa, about as accurately as persistence (2.1 and 1.8 Pa). In the last three fifths, when the seasonal baseline

Figure 7.3: Measurements and six-hour forecasts for three sols of the test period



Each forecast is drawn at the time for which it is valid. GBDT: mean of the forecasts of seeds 1–5.
Source: own elaboration based on the stored forecasts of the final runs (Chapter 5); sols 1560–1562.

fell from 741 to 627 Pa, its error rose to 7.0–8.4 Pa. In the last fifth, 22 % of the pressures to be forecast lie below the lowest pressure of the whole training period. A tree ensemble is piecewise constant in its inputs and does not extrapolate beyond the targets on which it was trained[2]; the lowest 24-hour forecast of the GBDT in that fifth is 604 Pa, whereas the measurements reach 586 Pa. The importances agree: 69–71 % of the gain of every pressure model comes from the seasonal baseline (Table 7.3 shows two of the four), so the model anchors its forecast to a level that the climb has made invalid. This is a case of dataset shift between the training and the test period[3], caused here by the motion of the station and not by the weather.

Third, the same model class beats persistence once the target no longer contains the level. A first indication came from the cross-site experiments, which forecast the change of pressure over the horizon on the hourly grid (Section 7.3): there the GBDT trained and tested at Jezero has errors of 1.22, 1.92, 1.22 and 1.29 Pa at 1, 6, 24 and 48 Mars hours, against 2.80, 8.61, 1.73 and 1.85 Pa for persistence. Because those experiments differ from the single-site ones in grid, horizon unit and inputs, a like-for-like check was added after the results above were known. The GBDT of RQ1 was trained again, with the same table, features, periods, settings, seeds

[2]Cf. T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd ed. Springer Series in Statistics. New York: Springer, 2009.

[3]Cf. J. Quiñonero-Candela et al., eds. *Dataset Shift in Machine Learning*. Neural Information Processing Series. Cambridge, MA: MIT Press, 2009.

Table 7.4: Bias and dispersion of the test errors

		Persistence		GBDT		TCN	
Target	H (h)	Bias	Dispersion	Bias	Dispersion	Bias	Dispersion
Air temperature (K)	1	+0.11	5.35	−0.08	2.09	+0.48	2.02
	6	+2.24	23.32	+0.23	2.94	+1.43	2.59
	24	+0.03	3.93	−0.16	2.01	+0.73	2.11
	48	+0.27	6.85	+0.19	2.26	+1.19	2.61
Pressure (Pa)	1	−0.09	2.72	+3.94	4.18	+1.76	2.77
	6	−0.32	8.69	+12.21	6.38	+9.36	3.97
	24	+0.47	2.09	+3.68	4.78	+7.38	3.30
	48	+0.83	3.76	+5.30	5.97	+9.64	3.79
Wind speed ($\mathrm{m\,s^{-1}}$)	1	−0.04	1.65	+0.23	1.27	+0.62	1.43
	6	−0.31	2.92	+0.33	1.33	+0.79	1.40
	24	+0.04	1.61	+0.23	1.22	+1.15	1.51
	48	+0.11	1.70	+0.26	1.21	+0.47	1.30

Bias: mean of forecast minus measurement; dispersion: standard deviation of the same differences; $\mathrm{RMSE}^2 = \mathrm{bias}^2 + \mathrm{dispersion}^2$ (Chapter 6). GBDT: mean over seeds 1–5.

Source: own elaboration based on the stored forecasts of the final runs (Chapter 5).

and test rows, and with one change: each booster was fitted to the change $v(t + H) - v(t)$ of Chapter 6, and $v(t)$ was added back before scoring. Table 7.5 shows the outcome. For pressure the error falls by 37–73 % relative to the GBDT on absolute values. The change-target GBDT beats persistence at 1, 24 and 48 hours, with errors of 1.74, 1.61 and 2.64 Pa and a skill of +0.36, +0.25 and +0.31, with all five seeds and with intervals well above zero, and it does so in every fifth of the test period at 1 and 24 hours. Its bias is below 0.7 Pa at these horizons. At six hours it only equals persistence (+0.00, interval −0.01 to +0.02); its errors there keep an offset of +4.2 Pa, so the six-hour change, which is mostly the diurnal tide, was not the same on the rim as in the training period.

The check confirms that the negative result of the declared design follows from forecasting the absolute pressure of a climbing rover against a fixed seasonal baseline, and not from a lack of predictability: stated as a change, pressure at Jezero can be forecast a quarter to a third more accurately than by persistence at 1, 24 and 48 hours. The change target is not a general improvement, however. For air temperature it raises the error of the GBDT by 5–17 %, and for wind it changes little; an absolute target lets the trees place the forecast on the diurnal cycle directly, which serves a variable without a drifting level better. The check has two limits. It was decided after the test results of the declared design had been seen, so it is exploratory and does not replace those results, and it covers the GBDT only; the TCN was not retrained on changes.

7.1.4 Wind speed

The wind results rest on little data: derived wind exists for sols 36–312 only (Chapter 3), so the models were trained on about 6,000 to 9,700 rows and tested on 29 to 37 sols of one season.

Table 7.5: Test RMSE of the GBDT trained on absolute values and on changes over the horizon

Target	H (h)	Better reference	GBDT, absolute target	GBDT, change target	Difference (%)	Skill of the change-target GBDT against the better reference
Air temperature (K)	1	4.51	**2.10** ± 0.01	2.20 ± 0.04	+4.8	+0.51 (+0.50 to +0.52)
	6	4.47	**2.95** ± 0.04	3.34 ± 0.02	+13.4	+0.25 (+0.22 to +0.28)
	24	3.93	**2.02** ± 0.02	2.18 ± 0.01	+8.1	+0.45 (+0.42 to +0.47)
	48	4.50	**2.27** ± 0.01	2.65 ± 0.02	+16.9	+0.41 (+0.39 to +0.43)
Pressure (Pa)	1	2.72	5.75 ± 0.11	**1.74** ± 0.07	−69.7	+0.36 (+0.35 to +0.37)
	6	8.70	13.78 ± 0.84	**8.66** ± 0.27	−37.2	+0.00 (−0.01 to +0.02)
	24	2.14	6.03 ± 0.32	**1.61** ± 0.08	−73.3	+0.25 (+0.24 to +0.26)
	48	3.85	7.99 ± 0.79	**2.64** ± 0.08	−66.9	+0.31 (+0.30 to +0.33)
Wind speed ($m\,s^{-1}$)	1	1.33	**1.29** ± 0.01	1.37 ± 0.01	+6.7	−0.04 (−0.07 to +0.00)
	6	1.33	1.37 ± 0.01	1.38 ± 0.01	+0.6	−0.03 (−0.08 to +0.01)
	24	1.26	**1.25** ± 0.01	1.30 ± 0.01	+4.0	−0.03 (−0.08 to +0.02)
	48	1.24	1.24 ± 0.02	1.24 ± 0.01	−0.1	−0.00 (−0.07 to +0.07)

Mean and standard deviation over seeds 1–5; the same 236,127 test forecasts as in Table 7.1. Better reference: the more accurate of persistence and the typical day. Difference: change target relative to absolute target. In parentheses: 95 % sol-block interval. Bold: the lower error where the interval of the difference excludes zero.

Source: own elaboration based on the stored forecasts of the runs of 27 October and 12 November 2026 (Chapter 5).

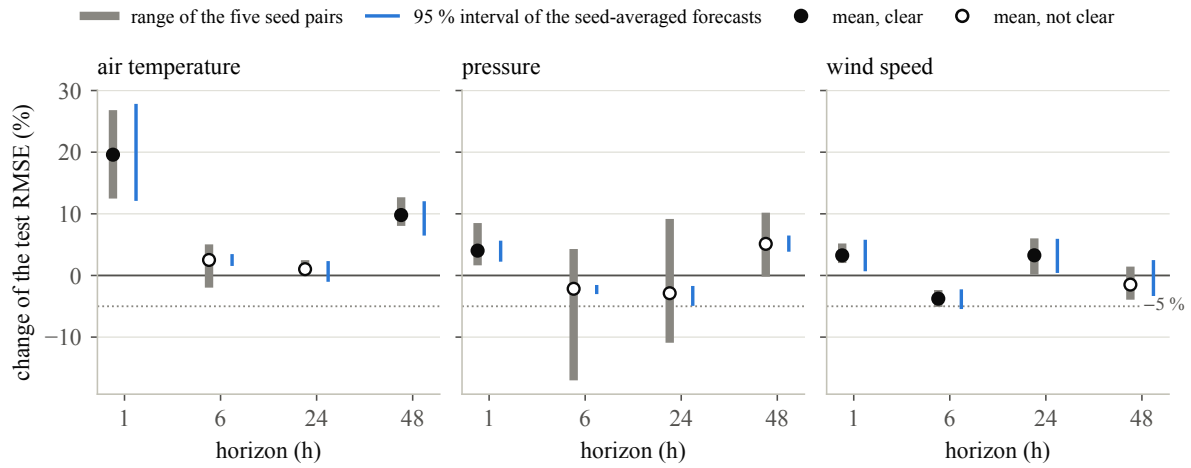
Within this window the GBDT reduces the error of persistence by 22–53 %, but it does not improve on the typical day: its skill against that reference is +0.03, −0.03, +0.01 and 0.00, and the intervals of the differences include zero at 6, 24 and 48 hours. The importances explain the tie. Only 12–17 % of the gain of the wind models comes from the history of the wind itself, and the rest from the time of sol, the solar geometry and the air temperature (Table 7.3 lists the largest shares). The GBDT has, in effect, relearned the mean diurnal cycle of the wind. Beyond that cycle, the 5-minute wind at Jezero behaved as noise for these inputs, which is consistent with the turbulent and vortex-rich daytime regime observed there (Chapter 2).

The TCN is worse than the GBDT and the typical day at every horizon and worse than persistence at 24 hours. Its forecasts are too high by 0.5–1.2 $m\,s^{-1}$ (Table 7.4), and the bias grows through the test window, from +0.2 to +1.7 $m\,s^{-1}$ between its first and last fifth at 24 hours. With an order of magnitude fewer training windows than for the other targets, the network did not learn a usable representation, and the typical day is the forecast to prefer for wind.

7.1.5 Hypothesis H1 and the comparison of the learned models

Hypothesis H1 required a learned model to reduce the 24-hour RMSE by at least 20 % relative to persistence on two of the three targets. The GBDT reduces it by 48.7 % for air temperature (interval 46.9–50.5 %) and by 22.8 % for wind speed (18.6–26.5 %), and increases it by 182 % for pressure. By its registered criterion H1 is therefore met by the GBDT. Two qualifications apply. The interval for wind includes the threshold, and for wind the typical day achieves the same reduction without any learning, so the substantive support for H1 comes from air temperature

Figure 7.4: Change of the test RMSE of the GBDT when the secondary signals are added



Source: own elaboration based on the seed check stored with the runs (Chapter 5).

alone. The TCN meets the criterion only for air temperature (43.1 %); for wind its 24-hour error is 17 % above that of persistence. In the additional check with change targets (Table 7.5), the GBDT also reduces the 24-hour error of pressure, by 24.9 % (interval 23.7–26.0 %); this formulation was not part of the registered design, so it is reported next to the verdict and not as part of it.

Between the two learned models as designed, the GBDT is the more accurate or an equal choice in ten of the twelve cells, the exceptions being pressure at 1 and 6 hours, where both lose to persistence. It is also far cheaper: its twelve models train in about 40 s on a laptop processor, against 31 minutes on a GPU for the TCN (Chapter 5). Both models overfit the training period to a visible degree, the GBDT with training errors of 1.5–1.9 K against validation errors of 2.4–3.0 K for air temperature and the TCN with early validation minima (Chapter 6), and both are held in check by early stopping. The comparison is not symmetric, however. The GBDT ran with fixed settings and five seeds, the TCN with a small search on one cell and a single run, and the TCN received the raw sequence without the solar angles that dominate the short-horizon GBDT models. The result is thus that a compact TCN, as configured here, does not improve on boosted trees for a single site; it is not a general statement about sequence models.

7.2 Value of the secondary signals

RQ2 was tested with the GBDT trained twice, on the 77 primary features and on the 132 features that add the water-vapor mixing ratio, the temperature of the humidity sensor, the two longwave fluxes and the solar radiation, with five seeds each (Chapter 6). Table 7.6 and Figure 7.4 show the change of the test RMSE caused by the secondary signals. No target shows a clear gain of at least 5 % at any horizon, so hypothesis H2 is not met for air temperature, pressure or wind speed.

The only clear gain is for wind at six hours, −3.8 %, below the threshold. Five cells show a clear loss, the largest for air temperature at one hour, where the error rises by 19.6 %, from

Table 7.6: Effect of the secondary signals on the test RMSE of the GBDT

Target	H (h)	Primary inputs	With secondary signals	Change (%)	Range of the seed pairs (%)	95 % interval (%)	Clear change
Air temperature (K)	1	2.10 ± 0.01	2.51 ± 0.10	+19.6	+12.5 to +26.8	+12.1 to +27.8	worse
	6	2.95 ± 0.04	3.02 ± 0.05	+2.5	−2.0 to +5.0	+1.5 to +3.5	—
	24	2.02 ± 0.02	2.04 ± 0.02	+1.0	+0.2 to +2.5	−1.0 to +2.3	—
	48	2.27 ± 0.01	2.49 ± 0.03	+9.8	+8.0 to +12.7	+6.5 to +12.0	worse
Pressure (Pa)	1	5.75 ± 0.11	5.98 ± 0.08	+4.0	+1.6 to +8.5	+2.2 to +5.6	worse
	6	13.78 ± 0.84	13.49 ± 0.74	−2.2	−17.0 to +4.3	−3.0 to −1.5	—
	24	6.03 ± 0.32	5.86 ± 0.34	−2.9	−10.9 to +9.2	−4.9 to −1.7	—
	48	7.99 ± 0.79	8.40 ± 0.63	+5.1	−0.2 to +10.2	+3.9 to +6.5	—
Wind speed ($m\,s^{-1}$)	1	1.29 ± 0.01	1.33 ± 0.02	+3.3	+2.0 to +5.2	+0.7 to +5.8	worse
	6	1.37 ± 0.01	1.32 ± 0.02	−3.8	−5.1 to −2.4	−5.4 to −2.3	better
	24	1.25 ± 0.01	1.29 ± 0.03	+3.3	+0.1 to +6.0	+0.4 to +5.9	worse
	48	1.24 ± 0.02	1.22 ± 0.01	−1.5	−3.9 to +1.4	−5.1 to +2.5	—

RMSE: mean and standard deviation over seeds 1–5. Change: of the mean RMSE; positive values mean that the secondary signals increased the error. Interval: sol-block interval of the difference between the seed-averaged forecasts, relative to the RMSE of the seed-averaged primary forecast. Clear change: all five seed pairs agree in sign and the interval excludes zero (Chapter 6); —: not clear.

Source: own elaboration based on the seed check stored with the runs (Chapter 5).

2.10 to 2.51 K. For pressure the mean changes are small and of mixed sign, and the individual seed pairs disagree: at six hours they range from −17.0 to +4.3 %. This spread is the reason for the five-seed protocol. A single pair of runs with the default seeds of the library had shown significant gains of 8.5 and 5.9 % for pressure at 6 and 48 hours, which the criterion as first formulated would have counted as support for H2. The first value lies inside the spread of the seed pairs, and at 48 hours four of the five seed pairs show a loss. The sol-block interval alone would have misled, because it covers the sampling of the test sols and not the variation of training.

The loss for air temperature was examined further, and it has an identifiable cause. With the secondary signals available, the one-hour model takes 73 % of its gain from the current upward longwave flux, in place of the solar zenith angle that dominates the primary model (Table 7.3). The upward flux measures the thermal emission of the ground, which is closely related to the air temperature (Chapter 4), so the substitution is plausible on the training period. In the test period, however, the flux carried instrument artifacts in sols 1551–1601, with 1,589 test values above 500 $W\,m^{-2}$ that the cap could limit but not repair (Chapter 4). In these sols, which hold 21 % of the test forecasts, the secondary signals raise the one-hour error by 76 %, from 1.98 to 3.49 K; in all other test sols they raise it by 2.2 %, from 2.13 to 2.17 K. At 48 hours the corresponding changes are +19.5 and +6.5 %. Most of the damage is therefore a failure of robustness: the model had moved its weight to a channel that later malfunctioned. Even outside the affected sols, however, the secondary signals bring no gain for air temperature.

Three reasons make the absence of a gain plausible. The secondary channels follow the same diurnal heating as the air temperature and are closely related to it and to each other (Chapter 4),

so they add little that the primary history and the solar geometry do not already contain. They are also less available: the mixing ratio exists only at night, in 31 % of the intervals, and the solar radiation in 53 %. Finally, pressure is almost unrelated to the thermal group within a sol (Chapter 4), and the pressure models with secondary signals take less than 1 % of their gain from them. The answer to RQ2 is thus negative for the tested model and data: adding the five secondary channels as a block did not improve the forecasts of the primary variables, and it made the model vulnerable to a sensor fault. The test covers the GBDT only, since the TCN was not trained with secondary inputs (Chapter 6), and it does not cover dust opacity, which the pipeline does not ingest.

7.3 Transfer between sites

7.3.1 Accuracy at the training site

The cross-site experiments use the hourly grid, change targets and the 15 relative inputs of Chapter 6. Table 7.7 first shows how the GBDT performs when trained and tested at the same site. At 1 and 6 Mars hours it is clearly skillful at all three stations and for both variables, with skill against persistence of +0.56 to +0.92. As in the single-site experiments, persistence is weak at six hours, with errors of 24–37 K and 9–33 Pa, and the climatology of the change is the harder reference there in every cell but pressure at Jezero; the GBDT beats the climatology as well, with errors of 2.8–3.0 K against 6.3–7.7 K and of 1.9–3.5 Pa against 6.7–11.7 Pa. At 24 and 48 hours the mean change over a whole number of sols is close to zero, so the climatology coincides with persistence, and the skill of the GBDT is smaller: +0.14 to +0.31 at Jezero and Gale, apart from air temperature at 48 hours at Jezero (+0.06), and at most +0.07 at Elysium Planitia, where two of the four intervals include zero. The shorter APSS record, with about 10,000 to 12,000 training rows, and the high repeatability of the weather at that site leave little for a model to add at one and two sols. Only the RMSE of the cross-site experiments is shown here. The MAE, stored with the run, places the GBDT on the same side of persistence in 95 of the 96 in-site and transferred cells; the exception is a transferred cell whose skill interval includes zero.

7.3.2 Transfer gaps

Figure 7.5 shows the transfer gap of Chapter 6, the relative increase of the error when the model comes from another site or from the two other sites pooled, for all 72 transferred cells. A model loses accuracy when it is moved in 69 of the 72 cells; in two cells the interval of the gap includes zero and in one the transferred model is slightly better. In 31 cells the lower bound of the interval exceeds 50 %, and these occur for both variables. Hypothesis H3, which required a gap above 50 % on at least one variable, is therefore supported.

The gaps depend strongly on the horizon. At 1 and 6 hours they range from +26 to +406 % for air temperature and from +36 to +960 % for pressure, with medians of +91 and +116 %, and 26 of these 36 cells exceed 50 % with their whole interval. At 24 and 48 hours the medians are

Table 7.7: Test RMSE and skill of the GBDT trained and tested at the same site on the hourly grid

		Jezero (MEDA)			Gale (REMS)			Elysium Planitia (APSS)		
Variable	H (h)	Pers.	GBDT	Skill	Pers.	GBDT	Skill	Pers.	GBDT	Skill
Air temperature (K)	1	5.31	2.07	+0.61	6.16	2.70	+0.56	7.86	1.43	+0.82
	6	24.31	2.79	+0.89	24.85	2.97	+0.88	36.85	2.90	+0.92
	24	2.89	2.07	+0.28	2.96	2.55	+0.14	1.28	1.29	−0.01*
	48	2.18	2.05	+0.06	3.26	2.74	+0.16	1.66	1.55	+0.07
Pressure (Pa)	1	2.80	1.22	+0.56	7.87	2.21	+0.72	5.86	1.58	+0.73
	6	8.61	1.92	+0.78	33.18	3.49	+0.89	18.11	3.02	+0.83
	24	1.73	1.22	+0.29	3.24	2.69	+0.17	2.02	1.99	+0.01*
	48	1.85	1.29	+0.31	3.67	3.00	+0.18	2.20	2.07	+0.06

H in Mars hours. Pers.: persistence. Skill against persistence; * the 95 % interval includes zero. Half widths of the skill intervals: at most 0.06. Test rows: 4,506–4,774 (Jezero), 14,051–14,534 (Gale), 1,299–1,652 (Elysium Planitia).

Source: own elaboration based on the transfer run and its bootstrap intervals (Chapter 5).

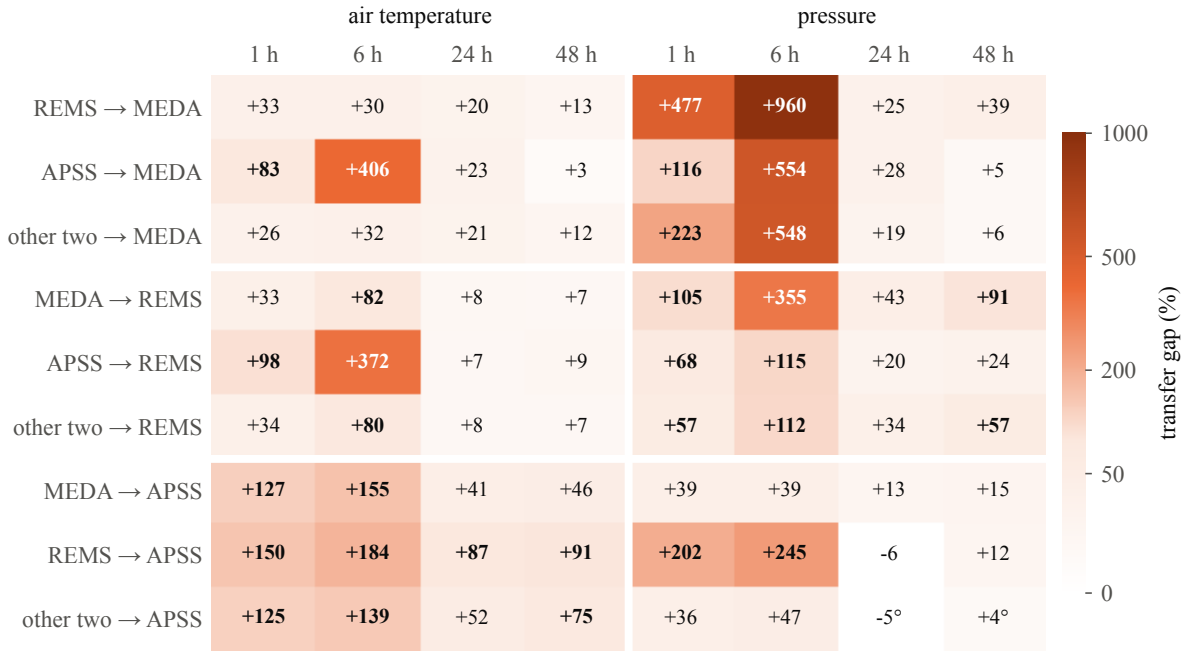
+16 and +19 % and only five cells exceed 50 %. The contrast follows from what the models learn at each horizon. At short horizons the forecast change is mostly the diurnal cycle of the site, which a model from elsewhere gets wrong; at one and two sols the change is small everywhere, the in-site models themselves have little skill (Table 7.7), and there is less to lose. The three cells without a loss are of this kind: pressure at Elysium Planitia at 24 and 48 hours, where the in-site model has a skill of +0.01 and +0.06.

The worst cells are those of pressure at Jezero: every model moved there has a gap of +548 to +960 % at six hours, and the model from Gale +477 % already at one hour. They are followed by air temperature moved from Elysium Planitia to the two crater sites at six hours, +372 and +406 %. The best transfers at short horizons are air temperature between Gale and Jezero (+30 to +33 % from Gale to Jezero) and pressure from Jezero to Elysium Planitia (+39 %). A large gap does not always mean a useless forecast. For air temperature at 1 and 6 hours, every transferred model is still better than persistence, with a skill of +0.13 to +0.85, so part of what the models learned applies at every site. For pressure, in contrast, 5 of the 18 transferred models at these horizons are worse than persistence, all of them tested at Jezero; the model from Gale reaches a skill of −1.52 and −1.36 there, a case of negative transfer[4].

Pooling the two other sites does not close the gap. The pooled model usually lies between the two single-site models or close to the better one, for example +26 % for air temperature at Jezero at one hour against +33 % from Gale and +83 % from Elysium Planitia, and +57 % for pressure at Gale at one hour against +68 and +105 %. In no cell at 1 or 6 hours does it remove the loss. With two training sites, a model cannot learn how the response depends on the site, because it receives nothing that describes the site.

[4]Cf. S. J. Pan and Q. Yang. “A Survey on Transfer Learning”. In: *IEEE Transactions on Knowledge and Data Engineering* 22.10 (2010), pp. 1345–1359.

Figure 7.5: Transfer gap of the GBDT for every pair of training and test site (%)



Rows: training data → test site; "other two": the two other sites pooled. Bold: the lower bound of the 95 % interval exceeds 50 %; °: the interval includes zero. Horizons in Mars hours.

Source: own elaboration based on the transfer run and its bootstrap intervals (Chapter 5).

7.3.3 Local and global structure

The second part of RQ3 asks what the transfer error reveals about local and global weather structure. The diurnal-cycle diagnostic of Chapter 6 gives a consistent picture. The *timing* of the mean diurnal cycle is nearly the same at the three sites: the cycles of any two sites correlate with $r = 0.93$–0.99 for air temperature and $r = 0.86$–0.96 for pressure. The *amplitude* is local: the mean diurnal range of air temperature is 55.9 K at Jezero, 51.7 K at Gale and 69.5 K at Elysium Planitia, and that of pressure 24.4, 75.5 and 39.6 Pa (Chapter 4). A model trained on the changes of one site has learned that site's amplitude. Moved from Gale to Jezero, where the diurnal pressure range is a third as large, it forecasts changes about three times too large, which is worse than forecasting no change at all; in the opposite direction it forecasts changes that are too small, and at 1 and 6 hours the gap is large but the model stays better than persistence. Between Gale and Jezero the ranges of air temperature differ by only 8 %, and the gaps from Gale to Jezero are the smallest at the short horizons.

The amplitude mismatch explains the gaps only in part. Over the 24 single-site transfers at 1 and 6 hours, the gap increases with the mismatch of the diurnal ranges, with a rank correlation of 0.50, but the air-temperature models from Elysium Planitia fail badly at six hours although their range differs from that of the crater sites by only 24–34 %. Differences in the shape of the cycle,

and at Elysium Planitia the thermal influence of the lander on the measured air temperature[5], are likely contributors, but the experiment cannot separate them. The reading that the results support is therefore the following. The part of the weather that is global at these horizons is the timing imposed by the Sun, and it transfers: it is why transferred air-temperature models still beat persistence. The part that is local is the amplitude of the response, which depends on topography, latitude and surface properties[6], and at Gale on the hydrostatic adjustment over the relief of the crater region[7]; it does not transfer, and it accounts for the largest gaps.

7.4 Discussion

7.4.1 Answers to the research questions

Table 7.8 collects the outcomes. For RQ1, the accuracy that machine learning adds depends on the variable. Air temperature can be forecast 1 to 48 hours ahead with an error of 2.0–3.0 K and a skill of +0.34 to +0.53 against the better reference. Pressure, forecast as an absolute value, is forecast best by persistence; the learned models fail for a reason that lies in this formulation and not in the variable, and a GBDT trained on the change of pressure beats persistence by 25–36 % at 1, 24 and 48 hours and equals it at 6 hours. Wind speed is forecast no better than by its mean diurnal cycle. For RQ2, the secondary signals did not measurably improve any primary forecast and harmed some. For RQ3, a model does not generalize to another site without loss; the loss is large at short horizons and small at one and two sols, and its pattern separates the shared timing of the diurnal cycle from its local amplitude.

The six objectives of Chapter 1 were carried out: the pipeline, the two references, the two learned models, the evaluation at four horizons with RMSE, MAE and skill, the test of the secondary signals and the transfer matrix all exist and are reported here. The aim, to determine for which variables and horizons the pipeline outperforms simple reference forecasts, is answered with a narrower positive result than the hypotheses anticipated: air temperature at all horizons and, once it is stated as a change, pressure at three of four horizons at one site, and the changes of both variables at short horizons at all three sites. That two of the three research questions end with largely negative answers is itself a finding of the evaluation practice adopted in Chapter 2, since every one of these conclusions depends on a reference forecast.

[5]Cf. D. Viúdez-Moreiras et al. “Effects of a Large Dust Storm in the Near-Surface Atmosphere as Measured by InSight in Elysium Planitia, Mars. Comparison With Contemporaneous Measurements by Mars Science Laboratory”. In: *Journal of Geophysical Research: Planets* 125.9, e2020JE006493 (2020).

[6]Cf. G. M. Martínez et al. “The Modern Near-Surface Martian Climate: A Review of In-situ Meteorological Data from Viking to Curiosity”. In: *Space Science Reviews* 212.1–2 (2017), pp. 295–338.

[7]Cf. M. I. Richardson and C. E. Newman. “On the relationship between surface pressure, terrain elevation, and air temperature. Part I: The large diurnal surface pressure range at Gale Crater, Mars and its origin due to lateral hydrostatic adjustment”. In: *Planetary and Space Science* 164 (2018), pp. 132–157.

Table 7.8: Outcomes of the three experiments against the hypotheses

	Criterion (Chapter 6)	Result	Outcome
H1	24-hour RMSE at least 20 % below persistence on two of three targets	GBDT: air temperature −48.7 %, wind speed −22.8 %, pressure +182 %; TCN: −43.1, +17 and +277 %	met by the GBDT; for wind the typical day does as well; with a change target (post hoc) pressure −24.9 %
H2	clear gain of at least 5 % at two horizons from the secondary signals	no clear gain of 5 % at any horizon; one clear gain of 3.8 %, five clear losses of 3.3–19.6 %	not met for any target
H3	cross-site RMSE more than 50 % above the in-site RMSE on at least one variable	lower interval bound above 50 % in 31 of 72 cells, for both variables; up to +406 % (air temperature) and +960 % (pressure)	supported

Source: own elaboration.

7.4.2 Comparison with the literature

The closest precedent reported a coefficient of determination of 0.991 for next-sol air temperature at Gale and no reference forecast[8]. The present test period shows how little such a number says: at 24 hours, persistence alone reaches a squared correlation with the measurements of 0.95 for air temperature and of 0.998 for pressure, and the GBDT reaches 0.99 for air temperature. The gain of the model is visible only in the error, which it halves. This confirms the argument of Chapters 1 and 2 that correlation measures potential and not actual skill[9], and the dependence of the skill on the reference found here, +0.87 or +0.34 for the same forecasts, illustrates Murphy's recommendation[10]. Among the few Martian studies that report skill, a conference abstract gave 18 % against persistence for column dust opacity one sol ahead[11]; the 49 % found here for air temperature refers to a different and more regular quantity.

The ranking of the models agrees with the terrestrial evidence reviewed in Chapter 2. Boosted trees on engineered features matched or beat the deep model on a data set of moderate size, as tree ensembles did on tabular benchmarks without time series[12], and simple references were not beaten on two of three targets in the declared design, as most teams of the M5 competition failed to beat the best simple benchmark[13]. The TCN, which was among the best models on a

[8] Cf. A. de Cabo García et al. "Prediction of Mars meteorological variables using artificial neural networks". In: *DYNA New Technologies* 9.1 (2022).

[9] Cf. A. H. Murphy. "Skill Scores Based on the Mean Square Error and Their Relationships to the Correlation Coefficient". In: *Monthly Weather Review* 116.12 (1988), pp. 2417–2424.

[10] Cf. *idem*, "Climatology, Persistence, and Their Linear Combination".

[11] Cf. L. Montabone et al. "Weather on Mars: From Monitoring to Forecasting". In: *Europlanet Science Congress 2026*. Conference abstract EPSC2026-1310. Copernicus, 2026.

[12] Cf. L. Grinsztajn, E. Oyallon, and G. Varoquaux. "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" In: *Advances in Neural Information Processing Systems 35 (Datasets and Benchmarks Track)*. 2022, pp. 507–520.

[13] Cf. S. Makridakis, E. Spiliotis, and V. Assimakopoulos. "M5 accuracy competition: Results, findings, and conclusions". In: *International Journal of Forecasting* 38.4 (2022), pp. 1346–1364.

Martian reanalysis one sol ahead[14], was not better than the GBDT at 24 hours on station data and lost to persistence for absolute pressure; that study reported no reference, so the two results do not conflict. The shortest horizon differs from terrestrial experience. At Austrian stations persistence was more accurate than neural networks for wind speed during the first two forecast hours[15]. At Jezero the GBDT beats persistence for wind speed already at one hour (+0.22), although the typical day does as well. For air temperature, a different variable, both learned models beat persistence at one hour with a skill of +0.61: the thin Martian atmosphere changes its temperature so quickly and so regularly that one hour of persistence already costs 5.3 K, which a model that knows the cycle avoids.

For RQ2, the earlier report that ground temperature, humidity and opacity did not improve a next-sol network[16] is consistent with the present result for the channels tested here, humidity, the longwave fluxes and solar radiation, with a different model, a different station, four horizons and a paired test; opacity was not tested. The finding that the secondary signals can harm through a sensor fault is new, as is the observation that a single pair of runs would have given the opposite verdict for pressure.

For RQ3, the transfer between the two Viking landers raised the errors of a next-sol network above 30 %, and its authors concluded that a network cannot be moved directly between regions[17]. The present study also finds a loss at one and two sols, with median gaps of +16 and +19 %, although the two measures are not the same; the much larger gaps at 1 and 6 hours could not be seen in a study with one horizon. Opacity forecasts transferred well between the Mars Exploration Rovers, in networks whose accuracy rose sharply once the previous sol's value was an input[18]; the small gaps found here occur likewise where persistence is strong. The preference for site-specific models agrees with station forecasting on Earth[19], and the failure of pooling agrees with the conditions under which global models succeeded there: hundreds of sites and static descriptors of each one[20], neither of which three Martian stations provide. Finally, the local amplitudes that the transfer error exposes are those documented by the observational studies: the large diurnal pressure range of Gale and the weaker tides of Jezero[21].

[14]Cf. S. Uprety et al. "Weather Prediction on Mars as a Multivariate Time Series Forecasting Problem". In: *Proceedings of the Workshop on AI-driven Data Engineering and Reusability for Earth and Space Sciences (DARES 2025)*. Vol. 4128. CEUR Workshop Proceedings. 2025, pp. 31–43. URL: `https://ceur-ws.org/Vol-4128/paper2.pdf` (accessed 2026-10-12).

[15]Cf. P. Papazek et al. "Feature selection, ensemble learning, and artificial neural networks for short-range wind speed forecasts". In: *Meteorologische Zeitschrift* 29.4 (2020), pp. 307–322.

[16]Cf. de Cabo García et al., *op. cit.*

[17]Cf. *ibid.*

[18]Cf. D. Bellutta. *The Fog of War: A Machine Learning Approach to Forecasting Weather on Mars*. 2017. URL: `https://arxiv.org/abs/1706.08915` (accessed 2026-10-12).

[19]Cf. Papazek et al., *op. cit.*

[20]Cf. F. Kratzert et al. "Toward Improved Predictions in Ungauged Basins: Exploiting the Power of Machine Learning". In: *Water Resources Research* 55.12 (2019), pp. 11344–11354.

[21]Cf. Richardson and Newman, *op. cit.*; A. Sánchez-Lavega et al. "Mars 2020 Perseverance Rover Studies of the Martian Atmosphere Over Jezero From Pressure Measurements". In: *Journal of Geophysical Research: Planets* 128.1, e2022JE007480 (2023).

7.4.3 Limitations

The results hold within limits set by the data, the methods and the evaluation.

Data. The single-site results come from one station and one test period of about 222 sols, which the rover spent on the crater rim, about 600 m higher than in most of the training period. By the seasonal coordinate of Chapter 4 the test period spans roughly 33° to 157°, northern spring and summer, so the accuracy during the dusty season and during dust storms was not tested. Wind was evaluated on 29 to 37 sols of one early season, and no wind could be transferred, because no REMS wind passes quality control. The MEDA air temperature is the mean of five sensors at two heights without quality flags, the APSS air temperature is perturbed by the lander, and its accuracy flag was not applied (Chapter 3). The hourly means of the three stations rest on different numbers of samples, from about five minutes to a full hour, and six sols of APSS pressure are missing. All three sites lie at low latitudes, so the transfer results say nothing about polar or mid-latitude sites.

Methods. The GBDT ran with fixed settings that were never tuned, and the TCN was tuned with six settings on one cell and trained once per cell; its run-to-run variation of about 6 % was estimated from two runs, and a seed check of the TCN was not made. The TCN was not tested with secondary inputs, in the transfer experiment or with change targets. The single-site models of the declared design forecast absolute pressure without a correction for elevation, which the results show to be the wrong formulation for a moving station; the check with change targets that demonstrates this was added after those results were known. All forecasts are point forecasts; no prediction intervals were produced. The models were trained once and not updated as the test period advanced.

Evaluation. Each experiment uses one chronological division with one test period. The rolling-origin evaluation with periodic refits that the literature recommends[22] was planned and not implemented, so the stability of the results over several test periods is unknown; a model refitted during the test period would probably have adapted to the lower pressure on the rim. The bootstrap intervals cover the sampling of the test sols only. The variation of training was measured for the GBDT with five seeds and not for the TCN. The rule for H2 was tightened after single runs had proved unstable, though before the runs that it judges (Chapter 6), and the check with change targets, the analysis of the upward longwave flux and the slices of the test period are post hoc. The feature importances come from one seed and share their values among related features in proportions that depend on the sampling. Finally, the single-site and cross-site experiments differ in grid, horizon unit, target and inputs, so their numbers cannot be compared cell by cell.

[22] Cf. L. J. Tashman. “Out-of-sample tests of forecasting accuracy: an analysis and review”. In: *International Journal of Forecasting* 16.4 (2000), pp. 437–450.

7.4.4 Strengths and weaknesses of the solution

The main strength of the work is its evaluation. Every learned model is compared with two reference forecasts on identical test rows; differences are judged against the sampling of the test sols and, for the GBDT, against the variation of training; and negative results are reported with the same care as positive ones. This practice exposed several effects that a single accuracy number would have hidden: the weakness of persistence at six hours, the dependence of an RQ2 verdict on the random seed, the elevation shift behind the pressure errors, which a change target then removed, and the sensor fault behind the air-temperature losses. A second strength is the correction of the pipeline itself: the defects described in Chapters 4 and 6 were found, fixed, covered by tests and followed by a repetition of every experiment, so the reported numbers describe the corrected system. A third is economy. The best model for the main positive result trains in under a minute on a laptop, and the whole study runs on the two consumer machines of Chapter 5. Finally, the transfer matrix compares the three modern stations on a common grid and in terms of predictability, which no study reviewed in Chapter 2 had done.

The main weaknesses are the pressure formulation of the declared single-site design, the thin evidence for wind, the asymmetric treatment of the two learned models and the single test period. Each has a direct remedy: change targets, tested here for the GBDT only and after the fact, or a hydrostatic reduction of pressure to a reference elevation; a longer wind record or calibrated wind once it is released; a seed check and a wider search for the TCN, and a tuned GBDT; and a rolling-origin evaluation with refits. The vulnerability shown in RQ2 suggests a further one, a plausibility check on every input channel before it reaches a model. These points, together with probabilistic forecasts, are taken up as future work in Chapter 8.

7.5 Summary

At Jezero Crater, gradient-boosted trees forecast the air temperature 1 to 48 hours ahead with an error of 2.0–3.0 K, between a third and a half below the better of two reference forecasts, and a compact TCN matched them at 1 and 6 hours but not beyond. For absolute pressure, persistence remained the best forecast, with errors of 2–9 Pa; the learned models failed because the rover's climb had shifted the pressure away from the level learned in training. In a check added afterwards, a GBDT trained on the change of pressure beat persistence by 25–36 % at 1, 24 and 48 hours and equaled it at 6 hours. For wind speed, no model improved on the mean diurnal cycle. Hypothesis H1 was met by the GBDT, with substantive support from air temperature only. The secondary signals brought no clear gain of 5 % at any horizon and, through artifacts of the upward longwave flux, raised the one-hour error of air temperature by a fifth, so H2 was not met. Models moved between Jezero, Gale and Elysium Planitia lost accuracy in 69 of 72 cells, by more than 50 % in 31, which supports H3; pooling two sites did not help, and the pattern of the gaps shows that the timing of the diurnal cycle is shared between the sites while its amplitude is local. Chapter 8 draws the conclusions.

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